Introduction to Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

IoT stands for _____.

a. Internet of Tasks
b. Internet of Tuples
c. Internet of Things
d. None of these

Correct Answer: c. Internet of Things

Detailed Solution: The full form of IoT is "Internet of Things"

See lecture 1 (Introduction to IoT – Part - I) @ 1:30

QUESTION 2:

Which of the following technologies have unified and has resulted in the evolution of IoT?

a. High-power embedded systems
b. Super Computing
c. Engine Technology
d. None of these

Correct Answer: d. None of these

Detailed Solution: Unification of technologies which has resulted in the advancement of IoT are –

a. Low-power embedded systems

b. Cloud Computing

c. Big Data

d. Machine Learning

e. Networking

See lecture 1 (Introduction to IoT – Part - I) @ 5:54

QUESTION 3:

Which of the following are the enables of IoT?

a. RFID
b. Nanotechnology
c. Sensors
d. All of these

Correct Answer: d. All of these

Detailed Solution: The enables of IoT are –

a. RFID

b. Nanotechnology

c. Sensors

d. Smart Networks

See lecture 2 (Introduction to IoT – Part - I) @ 12:50

QUESTION 4:

Which of the following is NOT a function of an IoT Gateway?

a. Connects the IoT LAN to a WAN

b. Can implement several LAN and WAN

c. Both (a) and (b)

d. Neither (a) Nor (b)

Correct Answer: d. Neither (a) nor (b)

Detailed Solution: The functionalities of an IoT Gateway are –

a. Connects the IoT LAN to a WAN

b. Can implement several LAN and WAN

c. Forwards packets between LAN and WAN on the IP layer

See lecture 2 (Introduction to IoT – Part - II) @ 3:09

QUESTION 5:

State whether the following statement is True or False.

Statement: "The ITU vision is approaching reality as the present-day networked devices have outnumbered humans on earth."

a. True

b. False

Correct Answer: a. True

Detailed Solution: The ITU vision is approaching reality as the present-day networked

devices have outnumbered humans on earth.

See lecture 2 (Introduction to IoT – Part - II) @ 02:35

QUESTION 6:

State True or False.

Statement: "In Multi-homing, a node/network is connected to multiple networks for improved reliability.

a. True
b. False

Correct Answer: a. True

Detailed Solution: In multi-homing, a node/network is connected to multiple networks for improved reliability.

See lecture 2 (Introduction to IoT – Part - II) @ 15:22

QUESTION 7:

Which of the following are the features of an IoT Gateway?

a. A router connecting IoT LAN and WAN to Internet
b. Can implement several LAN and WAN
c. Forwards packets between LAN and WAN
d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are the features of an IoT gateway –

- A router connecting IoT LAN and WAN to the Internet
- Can implement several LAN and WAN

- Forwards packets between LAN and WAN

See lecture 2 (Introduction to IoT – Part - II) @ 04:00

QUESTION 8:

State True or False.

Statement: "IoT Proxy performs active application layer functions between IoT nodes and other entities."

a. True
b. False

Correct Answer: a. True

Detailed Solution: IoT Proxy performs active application layer functions between IoT nodes and other entities.

See lecture 2 (Introduction to IoT – Part - II) @ 04:00

QUESTION 9:

The Gateway has a _____ address prefix, which can be used to identify them globally.

a. Non-unique
b. Unique
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: b. Unique

Detailed Solution: The Gateway has a unique address prefix, which can be used to identify

them globally.

See lecture 2 (Introduction to IoT – Part - II) @ 07:00

QUESTION 10:

The IPv4 notation uses _____ number of bits to represent an address.

e. 33
f. 32
g. Both (a) and (b)
h. Neither (a) nor (b)

Correct Answer: b. 32

Detailed Solution: The IPv4 notation uses 32 bits to represent an address.

See lecture 2 (Introduction to IoT – Part - II) @ 16:33

QUESTION 11:

The “Destination Address” field in the IPv4 header is of _____ bits in length.

a. 32
b. 64
c. 128
d. None of these

Correct Answer: a. 32

Detailed Solution: The IPv4 uses 32 bits to represent the “Destination Address”.

See lecture 2 (Introduction to IoT – Part - II) @ 17:53

QUESTION 12:

We can sense temperature using _____?

a. An actuator

b. A Transducer

c. A processor

d. None of these

Correct Answer: d. None of these

Detailed Solution: We can sense temperature using a Sensor.

See lecture 3 (Sensing) @ 01:00

QUESTION 13:

State True or False.

Statement: "A sensor is not only sensitive to the measured property, but it also influences the measured property."

a. True

b. False

Correct Answer: b. False

Detailed Solution: A sensor is only sensitive to the measured property. It does not influence the measured property.

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See lecture 3 (Sensing) @ 12:11

QUESTION 14:

Based on the output, sensors are classified as _____.

a. Scalar and Vector

b. Analog and Digital

c. Both (a) and (b)

d. Neither (a) nor (b)

Correct Answer: b. Analog and Digital

Detailed Solution: Based on the output, sensors are classified as Digital and Analog.

See lecture 3 (Sensing) @ 13:05

QUESTION 15:

What is the full form of SMP in the context of IoT actuators?

a. Shape Master Polymers

b. Switch Mode Power

c. Sensor Making Polymer

d. None of these

Correct Answer: d. None of these

Detailed Solution: The full form of SMP is Shape Memory Polymers.

See Page number – 111, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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